1. Digital Forensics Basics

1. What is digital forensics?

Digital forensics is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence from electronic devices.

2. What are the four main stages of the forensic process?

1. Collection: Acquire digital evidence.
2. Examination: Extract relevant data from the evidence.
3. Analysis: Interpret the data to determine what happened.
4. Reporting: Document and present the findings.

3. What is chain of custody and why is it important?

Chain of custody is the documented record of everyone who handled the evidence, when they handled it, and what they did with it. It helps establish evidence authenticity and integrity.

4. Why must forensic examiners work on copies rather than the original device?

To prevent accidental modification or damage to the original evidence and preserve its integrity for investigation.

2. Evidence Preservation and Hashing

5. What is a forensic image and why is it created?

A forensic image is a bit-by-bit copy of a storage device, including allocated files, deleted data, and other recoverable information. It allows investigators to analyze evidence without altering the original.

6. How is a cryptographic hash used to verify evidence integrity?

A hash generates a unique-looking digital fingerprint of data. Investigators calculate the hash before and after copying or analysis. Matching hashes indicate that the data has not changed, assuming a secure algorithm.

7. Why is SHA-256 preferred over MD5 in modern forensics?

SHA-256 is more resistant to collisions, where different data produces the same hash. MD5 is considered cryptographically weak, so SHA-256 is preferred for stronger integrity verification.

8. What is a write blocker and when is it used?

A write blocker is a hardware or software tool that prevents data from being written to a storage device. It is used during evidence acquisition to avoid modifying the original evidence.

3. File Systems and Artifacts

9. What are MACB timestamps and what does each represent?

- M — Modified: When file content was last changed.
- A — Accessed: When the file was last accessed.
- C — Changed: When file metadata or file-system status changed.
- B — Birth: When the file was created.

Note: The exact meaning and availability of timestamps depend on the operating system and file system.

10. Where does deleted data go and why is it recoverable?

Deletion often marks the file's storage space as available for reuse rather than immediately erasing its contents. The data may be recoverable until it is overwritten or otherwise erased.

11. What is file carving?

File carving is the recovery of files from raw data by identifying file signatures and structures, often without relying on file-system metadata.

12. Give examples of common forensic artifacts on a Windows system.

- Windows Registry
- Event Logs
- Browser history and downloads
- Prefetch files
- Recycle Bin contents
- LNK shortcut files
- Jump Lists
- USB connection records

- Recently accessed files

4. Forensic Tools

13. What is Autopsy and what is it built on?

Autopsy is a graphical digital forensics platform used to analyze disk images and investigate digital evidence. It is built on The Sleuth Kit.

14. What kind of analysis does Autopsy automate through ingest modules?

Ingest modules can automate file hashing, keyword searching, file-type identification, deleted-file detection, metadata extraction, and the identification of selected known files or artifacts.

15. What information does ExifTool reveal about a file?

ExifTool extracts metadata such as the camera model, creation date, GPS coordinates, author, software used, image dimensions, and other file-specific properties, depending on the format.

16. What is The Sleuth Kit and what are some of its commands?

The Sleuth Kit is a collection of command-line tools and a library for analyzing disk images and file systems.

Examples:

- mmls — Displays disk partition layouts.
- fls — Lists files and directories, including deleted entries where available.
- icat — Extracts file contents using a file-system metadata address.
- istat — Displays metadata about a file-system object.
- fsstat — Displays file-system information.

17. What is file carving good for and which tools perform it?

File carving is useful for recovering deleted or lost files when file-system metadata is missing or damaged.

Common tools include PhotoRec, Foremost, and Scalpel.

18. Why is memory forensics with Volatility important?

Volatility analyzes RAM captures to identify running processes, network connections, loaded modules, and other volatile information. It can help investigate malware, suspicious activity, and fileless attacks.

5. Metadata and Reporting

19. Why is GPS metadata significant in an investigation?

GPS metadata can help establish where a photograph or other file was created, supporting location-based timelines and investigations.

20. What privacy risk does metadata create when files are shared publicly?

Metadata may expose personal information, such as a person's location, identity, device details, or the time a photo was taken.

21. What should a professional forensic report include?

- Executive summary
- Case information
- Scope and objectives
- Methodology and tools used
- Evidence collected and cryptographic hashes
- Findings and supporting evidence
- Conclusions and limitations
- Recommendations, when appropriate
- Examiner's name and report date

22. Why must a forensic report be objective and reproducible?

Objectivity prevents personal bias from influencing conclusions. Reproducibility allows another examiner to follow the documented methods and independently verify the findings.